

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	30 % of CIA	Total Marks:	20 (Answer ANY ONE)
CO3 Statement:	Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)		
Student Name:	Kodavati Jahnavi Sai Satya	Reg. No.:	192372312
Section / Batch:	Slot A	Date:	07-08-2026

Code of Conduct

I K Jahnavi - 192372312 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K Jahnavi

Instructions

- Duration: 10 minutes

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

1. A CDMA system requires $E_b/N_0 = 5$ dB. Derive the required SIR in dB for a processing gain of 128, using $SIR \approx E_b/N_0 - 10\log_{10}(PG)$, showing every step of the conversion and subtraction. Explain why this relationship between E_b/N_0 and required SIR is a central design target in CDMA engineering, and discuss what a lower required E_b/N_0 (via better coding) would mean for system capacity.



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CO3 - AT2 Technical Presentation

K Jahnavi - 192372312

ITA0301 Mobile Computing

Course Faculty : Dr. Terrance Frederick Fernandez,
Professor, SIMATS Engineering, SIMATS

PROBLEM

- A CDMA system requires $E_b/N_0 = 5$ dB. Derive the required SIR in dB for a processing gain of 128, using $SIR \approx E_b/N_0 - 10\log_{10}(PG)$, showing every step of the conversion and subtraction. Explain why this relationship between E_b/N_0 and required SIR is a central design target in CDMA engineering, and discuss what a lower required E_b/N_0 (via better coding) would mean for system capacity.

CDMA System Design: Relationship Between E_b/N_0 , SIR, and Processing Gain

What is CDMA ?

- CDMA :- Code Division Multiple Access
- Multiple users share the same frequency simultaneously.
- Each user is assigned a unique spreading code and receiver uses same code to recover correct signal.

Spreading code :

- A unique binary code assigned to each user.
- It spreads one data bit into many chips.

Processing Gain (PG) :

$$PG = \frac{\text{Chip Rate}}{\text{Data Rate}}$$

E_b/N_0 & SIR Ratio

E_b/N_0 :

Energy per Bit (E_b)

Noise Power (N_0)

- Measures signal quality.
- Higher $E_b/N_0 \rightarrow$ Fewer errors.

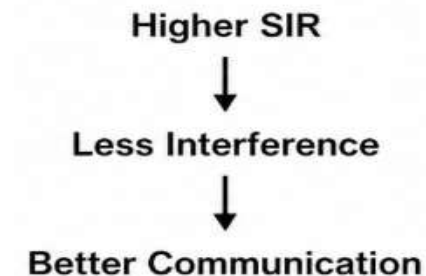


SIR :

Signal Power (S)

Interference Power (I)

- Compares signal with interference
- Interference : Unwanted noise from other CDMA users.



SIR Required Calculation :

Given

$$E_b/N_0 = 5 \text{ dB}$$

$$\text{Processing Gain (PG)} = 128$$

Step 1: Convert Processing Gain in dB

$$10 \log_{10}(PG)$$

$$10 \log_{10}(128) = 21.07 \text{ dB}$$

Step 2: Calculate SIR

$$SIR = E_b/N_0 - 10 \log_{10}(PG)$$

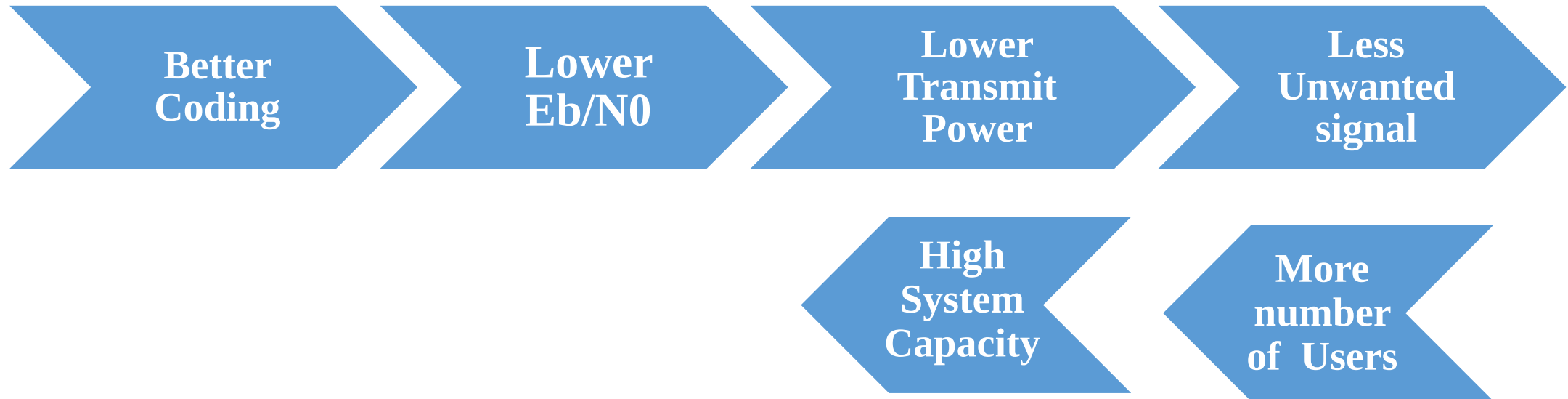
$$SIR = 5 - 21.07$$

$$SIR = -16.07 \text{ dB}$$

- ✓ **Required SIR** = -16.07 dB
- ✓ Negative SIR is normal in CDMA due to Processing gain.

Effect of Lower E_b/N_0 :

- Less signal energy is required.
- Users transmit with lower power.
- Interference decreases.
- More users can share the network.



Conclusion

- The required SIR is **-16.07 dB** for $E_b/N_0 = 5$ dB and Processing Gain = 128.
- A negative SIR is normal in CDMA because the spreading code and processing gain help recover the signal even below the interference level.
- This relationship helps engineers design reliable and high-capacity CDMA systems.



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THANK YOU

